Rational Functions and Asymptotes

ClipSAT Printable Practice Suite (clipsat.org) • Topic 8.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Finding asymptotes and intercepts

For $f(x) = \frac{x+2}{x-3}$, find:

a the vertical asymptote

b the horizontal asymptote

c the x-intercept

d the y-intercept

2 [Foundational] • Target: Finding asymptotes and intercepts

Find all asymptotes of $f(x) = \frac{2x^2+1}{x^2-4}$.

3 [Foundational] • Target: Finding asymptotes and intercepts

Under the structural constraint that consider $f(x) = \frac{x^2-9}{x-3}$.

a Simplify f and identify the feature at $x = 3$.

b Does the graph have a vertical asymptote?

4 [Foundational] • Target: Behaviour near asymptotes

For $f(x) = \frac{x+2}{x-3}$, describe the behaviour of $f(x)$ as:

a $x \to 3^+$ **b** $x \to 3^-$ **c** $x \to \infty$

5 [Foundational] • Target: Behaviour near asymptotes

Write a rational function with vertical asymptotes at $x = 1$ and $x = -4$ and horizontal asymptote $y = 0$.

6 [Foundational] • Target: Classifying end behaviour from degree

Without dividing, state the horizontal or slant asymptote (or neither) for each function based on the degrees of numerator and denominator:

a $f(x) = \frac{3x+1}{x^2-4}$ (degree 1 over 2)

b $f(x) = \frac{5x^2-1}{2x^2+3}$ (degree 2 over 2)

c $f(x) = \frac{x^3+2}{x^2-1}$ (degree 3 over 2)

7 [Foundational] • Target: Classifying end behaviour from degree

Find the slant asymptote of $f(x) = \frac{2x^2-3x+1}{x+2}$.

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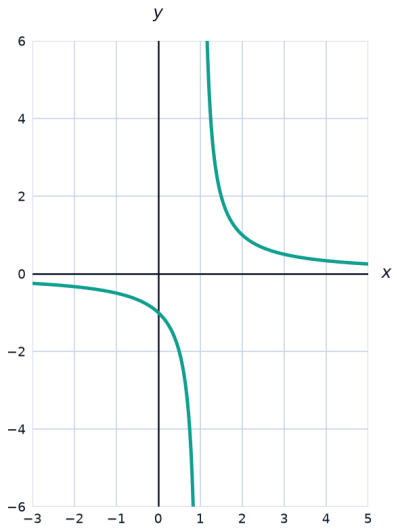
Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Holes versus asymptotes**

Construct a rational function that has a hole at $x = 3$ and a vertical asymptote at $x = -1$.

9 **[Intermediate]** • **Target: Reading asymptotic behavior from a graph**

The graph of $f(x) = \frac{1}{x-1}$ is shown. Use it to describe the behavior near $x = 1$ and as $x \to \pm\infty$.

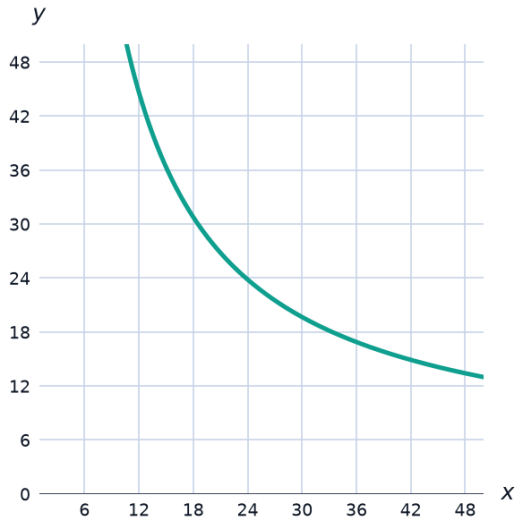


10 **[Intermediate]** • **Target: Domain restrictions and sign analysis**

Find the domain of $f(x) = \frac{x-1}{x^2-x-6}$, and determine the sign of $f(x)$ on the interval $(3, \infty)$.

11 **[Intermediate]** • **Target: Asymptotes in applied contexts**

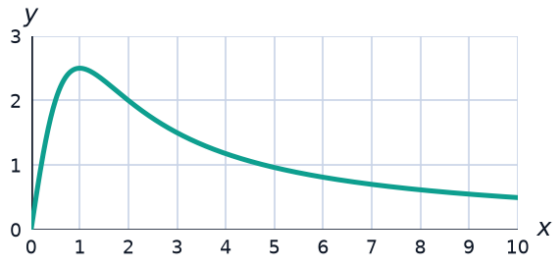
The average cost per unit of producing x items is $C(x) = \frac{500+3x}{x}$ (setup cost 500 riyals plus 3 riyals per unit). Find the horizontal asymptote and explain its meaning in context.



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12 [Intermediate] • Target: Asymptotes in applied contexts

A drug's concentration in the bloodstream is modeled by $C(t) = \frac{5t}{t^2+1}$ (mg/L, hours). Find the horizontal asymptote and interpret it, and find $C(0)$.

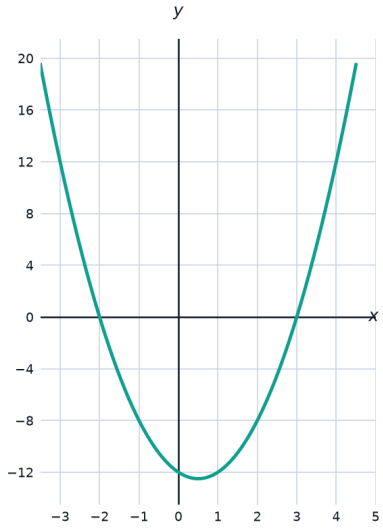


13 [Intermediate] • Target: Finding a rational function's equation from its features

Find the equation of a rational function with vertical asymptotes at $x = -2$ and $x = 3$, horizontal asymptote $y = 0$, and x-intercept at $x = 1$.

14 [Intermediate] • Target: Domain restrictions and sign analysis

For the rational function $r(x) = \frac{2x^2 - 2x - 12}{x^2 - 9}$, find: (a) the domain, (b) the coordinates of any removable discontinuity (hole), (c) the vertical and horizontal asymptotes, and (d) the x- and y-intercepts.



Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Asymptotes in applied contexts • Cross-Lesson: 3.03 Sum and Difference Formulas

Level 1 · FoundationsLet $q(x) = -(x+2)^2(x-1)(x-4)^3$. What is the degree of $q(x)$, does the graph cross or touch the x-axis at $x = -2$, and what is $\lim_{x \to \infty} q(x)$?

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16 **[Challenging]** • **Target: Finding a rational function's equation from its features** • **Cross-Lesson: 3.04 Double-Angle Formulas and Trigonometric Equations**

Let $f(x) = \frac{2x+1}{x-3}$ for $x \neq 3$. (a) Determine the inverse function $f^{-1}(x)$. (b) State the domain and range of $f^{-1}(x)$.

17 **[Challenging]** • **Target: Finding asymptotes and intercepts** • **Cross-Lesson: 4.01 Circles and Parabolas**

Let $f(x) = \frac{x^2-9}{x^2-5x+6}$. Identify the locations of all vertical asymptotes and removable discontinuities (holes) of $f(x)$.

18 **[Challenging]** • **Target: Behaviour near asymptotes** • **Cross-Lesson: 4.02 Ellipses**

The function f is defined by $f(x) = x^2 - 6x + 13$ for $x \geq 3$. (a) State the range of f. (b) Find an expression for $f^{-1}(x)$ and state the domain and range of f^{-1}.

19 **[Challenging]** • **Target: Classifying end behaviour from degree** • **Cross-Lesson: 4.03 Hyperbolas**

Find the equation of the slant (oblique) asymptote of the rational function $g(x) = \frac{3x^2-5x+7}{x-2}$, and compute $\lim_{x \to 2^+} g(x)$.

20 **[Challenging]** • **Target: Holes versus asymptotes** • **Cross-Lesson: 5.01 Vectors: Components and Operations**

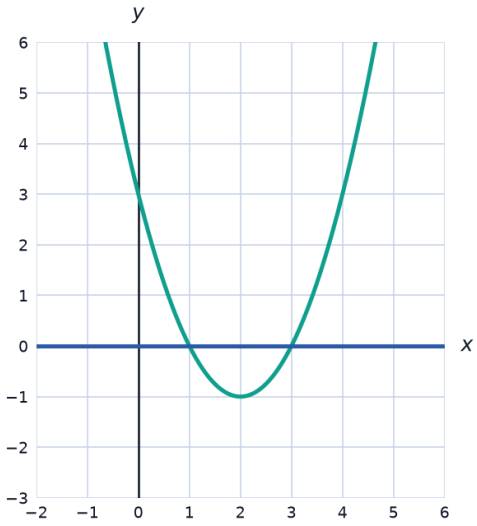
Let $f(x) = \frac{4e^{2x}}{e^{2x}+3}$. Prove that f is strictly one-to-one and find an explicit formula for its inverse function $f^{-1}(x)$, stating the domain of f^{-1}.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Finding asymptotes and intercepts

Consider the quadratic function $f(x) = x^2 - 4x + 3$. What are the x-intercepts of the graph of this function?



a. $(1, 0)$ and $(3, 0)$

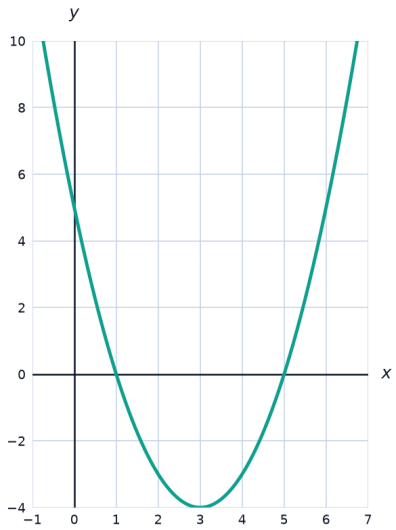
b. $(-1, 0)$ and $(-3, 0)$

c. $(1, 0)$ and $(-3, 0)$

d. $(-1, 0)$ and $(3, 0)$

22 [Foundational] • Target: Behaviour near asymptotes

Consider the quadratic function $f(x) = x^2 - 6x + 5$. What are the x-intercepts of the graph of $f(x)$?



a. x = 1 and x = 5

b. x = -1 and x = -5

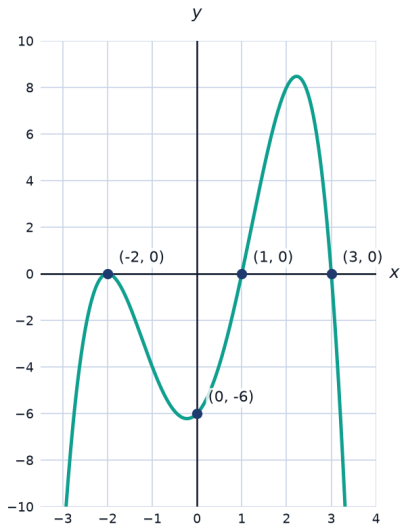
c. x = 1 and x = -5

d. x = -1 and x = 5

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23 [Foundational] • Target: Classifying end behaviour from degree

The graph of a polynomial function $f(x)$ of degree 4 is shown in the figure below. The graph is tangent to the x-axis at $(-2, 0)$, intersects the x-axis at $(1, 0)$ and $(3, 0)$, and crosses the y-axis at $(0, -6)$. A new polynomial $Q(x)$ is defined by $Q(x) = f(x) + \frac{1}{2}x^3$. When $Q(x)$ is divided by $x^2 - 4$, the remainder is the linear polynomial $R(x)$. What is the value of $R(5)$?



a. 24

b. 18

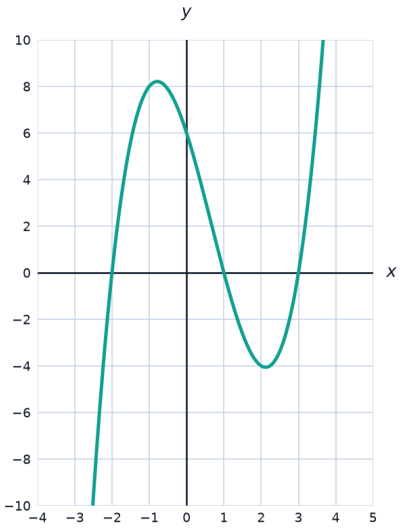
c. 17

d. 14

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24 [Foundational] • Target: Holes versus asymptotes

The graph of a polynomial function $f(x)$ is shown below. Which of the following could be the equation for $f(x)$?



a. $f(x) = (x + 2)(x - 1)(x - 3)$

b. $f(x) = (x - 2)(x + 1)(x + 3)$

c. $f(x) = -(x + 2)(x - 1)(x - 3)$

d. $f(x) = (x + 2)(x - 1)^2(x - 3)$

25 [Foundational] • Target: Reading asymptotic behavior from a graph

Which of the following statements describes the end behavior of the polynomial function $f(x) = -5x^4 + 2x^3 - 7x + 3$?

a. As $x \to -\infty$, $f(x) \to -\infty$, and as $x \to \infty$, $f(x) \to -\infty$

b. As $x \to -\infty$, $f(x) \to \infty$, and as $x \to \infty$, $f(x) \to -\infty$

c. As $x \to -\infty$, $f(x) \to -\infty$, and as $x \to \infty$, $f(x) \to \infty$

d. As $x \to -\infty$, $f(x) \to \infty$, and as $x \to \infty$, $f(x) \to \infty$

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26 [Foundational] • Target: Domain restrictions and sign analysis

For the polynomial $f(x) = 3x^2 + 2x - 1$, describe the end behavior of f.

a. down on the left, up on the right (as $x \to -\infty$, $f(x) \to -\infty$; as $x \to +\infty$, $f(x) \to +\infty$)

b. down on both ends (as $x \to -\infty$, $f(x) \to -\infty$; as $x \to +\infty$, $f(x) \to -\infty$)

c. up on the left, down on the right (as $x \to -\infty$, $f(x) \to +\infty$; as $x \to +\infty$, $f(x) \to -\infty$)

d. up on both ends (as $x \to -\infty$, $f(x) \to +\infty$; as $x \to +\infty$, $f(x) \to +\infty$)

27 [Foundational] • Target: Asymptotes in applied contexts

Find the x-intercepts and vertical asymptotes of $f(x) = \frac{x^2 - x - 6}{x^2 - 4}$.

a. x-int: $x = 3, -2$ (but $x = -2$ is a hole); VA: $x = 2$

b. x-int: $x = 3$; VA: $x = \pm 2$

c. x-int: $x = 3, -2$; VA: $x = \pm 2$

d. x-int: $x = 3$; VA: $x = 2$ only

28 [Foundational] • Target: Finding a rational function's equation from its features

Find the inverse of $f(x) = \frac{3x + 2}{x - 1}$ and state its domain.

a. $f^{-1}(x) = \frac{x + 2}{x - 3}$; domain: all reals except $x = 3$

b. $f^{-1}(x) = \frac{3x - 2}{x + 1}$

c. $f^{-1}(x) = \frac{x + 2}{x + 3}$

d. $f^{-1}(x) = \frac{x - 2}{x + 3}$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Finding asymptotes and intercepts

Find all intercepts and asymptotes of $f(x) = \frac{x^2 - x - 6}{x^2 - 4}$.

a. x-int $x = 3$; VA $x = 2$; slant asymptote

b. Only HA $y = 1$

c. x-int $x = 3,\ -2$; VA $x = \pm 2$; HA $y = 0$

d. x-int $x = 3$; hole at $x = -2$; VA $x = 2$; HA $y = 1$

30 [Intermediate] • Target: Behaviour near asymptotes

The reciprocal function $f(x) = \frac{1}{x}$ has asymptotes:

a. $x = 0$ and $y = 0$

b. No asymptotes

c. $x = -1$ and $y = -1$

d. $x = 1$ and $y = 1$

31 [Intermediate] • Target: Classifying end behaviour from degree

A polynomial of even degree with positive leading coefficient has end behavior:

a. $y \to -\infty$ as $x \to \pm\infty$

b. $y \to \infty$ as $x \to +\infty$ and $y \to -\infty$ as $x \to -\infty$

c. $y \to +\infty$ as $x \to \pm\infty$

d. Constant

32 [Intermediate] • Target: Holes versus asymptotes

What causes a hole in the graph of a rational function?

a. The y-intercept

b. A horizontal asymptote

c. A common factor in numerator and denominator

d. A vertical asymptote

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33 [Intermediate] • Target: Reading asymptotic behavior from a graph

If a rational function has no real zeros in its denominator, it has:

a. No horizontal asymptote

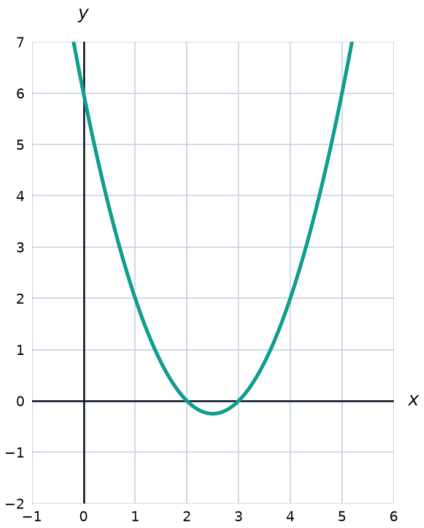
b. No x-intercepts

c. No vertical asymptotes

d. No range

34 [Intermediate] • Target: Domain restrictions and sign analysis

Consider the polynomial function $f(x) = x^2 - 5x + 6$. What are the roots of the function $f(x)$?



a. 1 and 2

b. 2 and 3

c. 3 and 4

d. 1 and 6

35 [Intermediate] • Target: Asymptotes in applied contexts

The graph of the rational function $g(x) = \frac{ax^2 + bx + 1}{x^2 - 4}$ has a horizontal asymptote at $y = 2$ and a vertical asymptote at $x = 2$. Given that $g(0) = -1/4$, find the value of a.

a. 1

b. 2

c. 3

d. 4

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36 [Intermediate] • Target: Finding a rational function's equation from its features

The function f is defined by $f(x) = \frac{ax+8}{cx+d}$ for all real numbers $xeq -\frac{d}{c}$, where a, c, and d are nonzero constants. In the standard (x, y) coordinate plane, the graph of $y = f(x)$ has a horizontal asymptote at $y = 2$, a vertical asymptote at $x = 3$, and an x-intercept at $(-4, 0)$. What is the value of $f^{-1}(4)$?

a. 10

b. 16

c. -2

d. 2

37 [Intermediate] • Target: Finding asymptotes and intercepts

The diagram shows the graph of the function $y = f(x)$. The function $f(x)$ is defined for all real numbers. Which of the following represents the graph of $y = f^{-1}(x)$?

a. Graph A

b. Graph B

c. Graph C

d. Graph D

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Behaviour near asymptotes • Cross-Lesson: 4.02 Ellipses

In the standard (x, y) coordinate plane, what is the equation of the horizontal asymptote of the graph of the function $f(x) = \frac{8x+3}{2x-5}$?

a. $y = 4$

b. $y = \frac{5}{2}$

c. $x = \frac{5}{2}$

d. $y = -\frac{3}{5}$

39 [Challenging] • Target: Classifying end behaviour from degree • Cross-Lesson: 4.03 Hyperbolas

For the rational function $f(x) = \frac{x-2}{x+3}$, what value of x is excluded from the domain?

a. x = -3

b. x = 2

c. x = 0

d. x = -2

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40 [Challenging] • Target: Holes versus asymptotes • Cross-Lesson: 5.01 Vectors: Components and Operations

Consider the polynomial function $f(x) = x^3 - 5x^2 + 2x + 8$. What is the value of $f(2)$?

a. -4

b. 0

c. 4

d. 8

41 [Challenging] • Target: Reading asymptotic behavior from a graph • Cross-Lesson: 5.02 The Dot Product

Determine if $f(x) = \frac{3x^2 - 5}{x^2 + 2}$ has any vertical asymptotes.

a. Yes at $x = 0$

b. Yes at $x = \pm\sqrt{2}$

c. Yes at $y = 3$

d. No (denominator $x^2 + 2 > 0$ always)

42 [Challenging] • Target: Domain restrictions and sign analysis • Cross-Lesson: 5.03 Parametric Equations

Find the cubic polynomial $P(x)$ with leading coefficient 2 whose graph passes through $(0, 6)$ and has zeros 1, -3 and $1 + i$. But wait — to have real coefficients, if $1 + i$ is a root, so is $1 - i$. That gives degree 4.

a. $P(x) = 2x^3 - 6$

b. $P(x) = 2(x - 1)(x + 3)(x - 1 - i)$

c. $P(x) = 2(x - 1)^2(x + 3)$

d. This is impossible for a cubic with real coefficients (need degree ≥4 for those zeros)

43 [Challenging] • Target: Asymptotes in applied contexts • Cross-Lesson: 6.01 Polar Coordinates and Conversions

Which of these is a rational function?

a. $f(x) = \sqrt{x}$

b. $f(x) = \frac{x^2 - 1}{x + 3}$

c. $f(x) = x^{2/3}$

d. $f(x) = 2^x$

44 [Challenging] • Target: Finding a rational function's equation from its features • Cross-Lesson: 6.02 Polar Equations and Graphs

Find the vertical asymptotes of $f(x) = \frac{x + 1}{x^2 - 4}$.

a. $x = \pm 2$

b. No VA

c. $x = -1, \pm 2$

d. $x = 2$ only

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45 [Challenging] • Target: Finding asymptotes and intercepts • Cross-Lesson: 7.01 Exponential Functions and Their Graphs

The domain of $f(x) = \frac{1}{x-3}$ excludes:

a. $x = 3$

b. $x = 0$

c. All reals

d. $x = -3$

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